Broyden’s Method

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Secant Condition:

$$f'\left(x^{(k)}\right) = \frac{f\left(x^{(k)}\right) - f\left(x^{(k-1)}\right)}{x^{(k)} - x^{(k-1)}}$$

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<u>Example 03-09</u>: Find root of the following system of equations using Broyden’s method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$
\begin{aligned}
x_1 + 2x_2 + x_3 + 4x_4 &= 20.700 \\
x_1^2 + 2x_1x_2 + x_4^3 &= 15.880 \\
x_1^3 + x_3^2 + x_4 &= 21.218 \\
3x_2 + x_3x_4 &= 21.100
\end{aligned}
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$$\mathbf{J}_{\partial\mathbf{f}/\partial\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2x_1 + 2x_2 & 2x_1 & 0 & 3x_4^2 \\ 3x_1^2 & 0 & 2x_3 & 1 \\ 0 & 3 & x_4 & x_3 \end{bmatrix}$$

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$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \mathbf{d}^{(k)}$$

$$\mathbf{J}_{\partial \mathbf{f}/\partial \mathbf{x}}^{(k)} = \mathbf{J}_{\partial \mathbf{f}/\partial \mathbf{x}}^{(k-1)} + \frac{\mathbf{f}\left(\mathbf{x}^{(k)}\right) \otimes \mathbf{d}^{(k)}}{\mathbf{d}^{(k)} \cdot \mathbf{d}^{(k)}}$$

$$\mathbf{J}_{\partial \mathbf{f}/\partial \mathbf{x}}^{(k)} \cdot \mathbf{d}^{(k+1)} = -\mathbf{f}\left(\mathbf{x}^{(k)}\right)$$

$$\vdots$$